

SLA COMPLIANCE & TICKET RESOLUTION DASHBOARD

IT Helpdesk Analytics in Power BI

Project Documentation
Prepared as a Data Analytics Portfolio Project
Tool: Microsoft Power BI Desktop

1. Project Overview

This project simulates a real-world IT Helpdesk / Support Ticketing environment and presents an interactive Power BI dashboard that tracks ticket volume, resolution performance, SLA (Service Level Agreement) compliance, and team workload. It is designed to demonstrate end-to-end BI development skills: data modeling, DAX calculations, interactive visuals, and drill-through navigation for root-cause analysis.

The dashboard answers questions a support operations manager would actually ask: How many tickets are open vs. closed? Which categories generate the most tickets? Who is carrying the heaviest workload? Are we meeting our SLA targets, and where are we breaching them?

2. Business Objective

The dashboard was built to help a support/IT operations team:

- Monitor overall ticket health (open, closed, and total volume) in real time.
- Identify which issue categories (Hardware, Software, Network, Security) drive the most support load.
- Evaluate individual and team workload distribution across support agents.
- Track SLA compliance and flag the proportion of tickets breaching resolution deadlines.
- Spot trends in ticket volume over time to support staffing and capacity planning.
- Drill through from summary KPIs into ticket-level detail for audit and root-cause investigation.

3. Dataset Description

The dataset used is a synthetic IT support ticketing log (ticketing_Dataset.csv) containing the following fields:

Column	Description
Ticket ID	Unique identifier for each support ticket (e.g., T001)
Date Opened	Date the ticket was logged
Date Closed	Date the ticket was resolved (blank if still open)
Resolution Due Date	SLA deadline by which the ticket should be resolved
Category	Issue type: Hardware, Software, Network, or Security
Assigned To	Support agent responsible for the ticket
Status	Current ticket state: Open or Closed

Dataset size: 1,557 ticket records spanning a full annual cycle (Jan–Dec).

4. Tools & Skills Demonstrated

Area	Details
BI Tool	Microsoft Power BI Desktop
Data Modeling	Star-schema style model with a dedicated Calendar (date) table for time intelligence
DAX	Custom measures for KPIs, percentages, SLA status, and ticket aging (see Section 6)
Visuals Used	KPI cards, donut chart, pie chart, clustered bar chart, line chart, slicers
Interactivity	Cross-filtering slicers (Status, Category, Assigned_To) and button-based drill-through

Area	Details
Navigation	Custom drill-through page with a back button for ticket-level detail views
Design	Consistent purple/navy brand theme, card-based KPI layout, clear visual hierarchy

5. Data Model

The model follows a simple star schema:

- Fact table: TicketingDB — one row per ticket, holding Ticket_ID, dates, Category, Assigned_To, and Status.
- Dimension table: Calendar Table — a standalone date table used to drive the "Tickets Over Time" trend and any future time-intelligence measures (MTD, YTD, month-over-month).
- Relationship: Calendar Table[Date] joined to TicketingDB[Date_Opened] (one-to-many).

Keeping a separate calendar table (rather than relying on the raw date column) is a best practice that enables consistent month/quarter/year grouping and protects against gaps in the date range.

6. Key DAX Measures

The following calculated measures power the KPI cards and visuals:

Measure	Purpose
Total Tickets	COUNTROWS of the ticket fact table — overall volume
Closed Tickets	Count of tickets where Status = "Closed"
Open Tickets	Count of tickets where Status = "Open"
Closed Tickets%	Closed Tickets ÷ Total Tickets
Open Tickets%	Open Tickets ÷ Total Tickets
SLA Days	Resolution Due Date – Date Opened, i.e. the SLA window allowed per ticket
SLA Status	Flags each ticket as "Within SLA" or "SLA Breach" by comparing Date_Closed (or today, for open tickets) against Resolution Due Date
Age Closed	Date Closed – Date Opened — actual days taken to resolve a closed ticket
Open Tickets Age	TODAY() – Date Opened — how long a still-open ticket has been outstanding

7. Dashboard Pages

Page 1 — Tickets Analysis (Landing Page)

The main analysis page gives a full operational snapshot:

- KPI strip: Total Tickets, Closed Tickets, Open Tickets, Closed %, Open % — an at-a-glance health check.
- Tickets by SLA Status (pie): share of tickets resolved within SLA vs. SLA breaches.
- Tickets by Category (bar): ticket volume by Hardware, Software, Network, and Security.
- Tickets by Assigned_To (clustered bar): total vs. open tickets per support agent — highlights workload and backlog by person.
- Tickets by Status (donut): overall open vs. closed split.
- Tickets Over Time (line): monthly ticket volume trend across the year, useful for spotting seasonal spikes.
- Slicer panel: filter the entire page by Status, Category, or Assigned_To.

Page 2 — Drill-Through: Tickets Details

A dedicated drill-through page lets a user right-click (or click) any Category or Assigned_To value on the main page and jump straight into the underlying ticket-level records:

- Ticket Details by Assigned_To: every ticket for the selected agent, with Category, Status, Open Ticket Age, and Total Tickets.
- Tickets Details by Category: every ticket for the selected category, with the same level of row-level detail.
- A back button returns the user to the main analysis page, preserving a smooth navigation flow.

This drill-through pattern demonstrates the ability to design multi-page, audit-ready dashboards rather than single flat reports — a key skill for real analyst roles where stakeholders need to move from "what happened" to "show me the tickets."

8. Key Insights

Based on the current dataset (1,557 tickets):

- **Overall health:** 1,380 tickets closed (88.6%) vs. 177 open (11.4%) — a healthy closure rate overall.
- **SLA risk:** The SLA Status split shows a large share of tickets breaching their resolution deadline, far outweighing tickets resolved within SLA — this is the single biggest opportunity area for the support team to address.
- **Category load:** Network issues are the top driver of ticket volume (636 tickets), followed by Security (419), with Hardware (252) and Software (250) roughly even and lowest.
- **Workload distribution:** Sarah Brown and Mike Johnson carry the largest ticket loads (331 and 330 respectively); David Wilson has a comparatively high number of open/backlogged tickets (40) relative to total volume, worth a closer look.
- **Volume trend:** Monthly ticket volume fluctuates through the year with a peak around March (~159 tickets) and a sharp dip in June (~101 tickets), suggesting a seasonal or event-driven pattern worth investigating further.

9. How to Use the Dashboard

- Use the slicers (Status, Category, Assigned_To) to filter every visual on the main page simultaneously.
- Hover over any chart for tooltip-level detail (exact counts, percentages).
- Click a bar in "Tickets by Category" or "Tickets by Assigned_To" and use the drill-through action to open the Ticket Details page for that specific slice.
- Use the back arrow on the drill-through page to return to the summary view.

10. Potential Enhancements

- Add a Resolution Time trend (average days-to-close) by month or by agent.
- Add a Top N Breached Tickets table sorted by days overdue, for immediate triage.
- Introduce a Priority field (High/Medium/Low) if available, to weight SLA analysis by severity.
- Publish to the Power BI Service and add a scheduled refresh if connected to a live ticketing system (e.g., Zendesk, ServiceNow, Jira Service Management).
- Add row-level security (RLS) so each agent only sees their own tickets, if this were used operationally.

11. Project Files

File	Description
ticketing_Dataset.csv	Source data (1,557 rows) used to build the model
Support_Tickets_Dashboard.pbix	Power BI report file containing the data model, DAX measures, and both report pages
Project Documentation (this file)	Overview, methodology, and insights for portfolio presentation

12. Suggested Dashboard Names

A few naming options considered for this project, ranked by how well they communicate business value to a portfolio reviewer:

- **SLA Compliance & Ticket Resolution Dashboard** — recommended. Leads with the business metric (SLA) hiring managers scan for.
- **IT Helpdesk Performance Tracker** — clean, functional, and easy to search/find.
- **TicketPulse: Support Operations Dashboard** — more brand-like, good if presenting as a product concept.
- **Helpdesk Analytics: Ticket Lifecycle & SLA Insights** — descriptive, strong for a portfolio website or resume bullet.
- **Support Ops Command Center** — enterprise, executive-facing tone.